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Control Barrier Function

Candidato:
Luca Mombelli
Matricola VR500265

Relatori:
Prof. Nicola Sanosetto

Correlatore:
Mauro Bonaifini

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Chapter 1

Control Barrier Function

1.1 Nagumo's Theorem

We presented Nagumo's Theorem following [3] , which is one of the fundamental building blocks for control barrier function

Definition 1.1.1. Let K be a subset of a finite dimensional vector-space (or even a normed space) X . We shall say that a function $x(\cdot) : [0, T] \to X$ is *viable* in K on $[0, T]$ if

$$\forall t \in [0, T] \ , \quad x(t) \in K$$

Consider

$$\begin{cases} \dot{x}(t) = f(x(t)) & \forall t \in [0, T] \\ x(0) = x_0 \end{cases} \tag{1.1.1}$$

con $f : \Omega \subset_{\text{op}} X \to X$

Definition 1.1.2. Let K be a subset of Ω. We shall say that K is *locally viable* under f (or enjoys the local viability propery for the map f) if for any initial state $x_0 \in K$, there exists $T > 0$ and a viable solution su $[0, T]$ to the differential equation 1.1.1 with initial condition x_0. It is said to be globally viable under f if we can always take $T = \infty$ The subset K is said to be invariant under f if for any initial state x_0 of K , all solutions to the differential equation 1.1.1 are viable in K

The concept of invariance depends upon the behavior of f on the domain Ω outside K . But we observe that viability property depends only on the behavior of f on K

Definition 1.1.3 (Bouligand Contingent cone)**.** Let X be a normed space , K be a nonempty subset of X and x belong to K . The contingent cone to K at x is the set

$$T_K(x) = \{v \in X \mid \liminf_{h \to 0^+} \frac{d_K(x + hv)}{h} = 0\}$$

where $d_k(x) := \inf_{y \in K} \|x - y\|$

In other worlds , v belongs to $T_K(x)$ if and only if exist a sequence of $h_n > 0$ $h_n \to 0^+$ and a sequence $v_n \in X$, $v_n \to v$ such that

$$\forall n \in \mathbb{N} \ , \ x + h_n v_n \in K$$

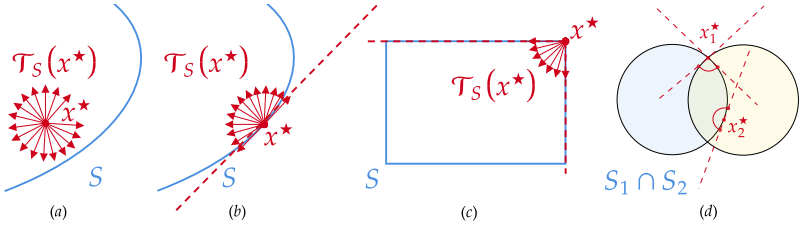


Figure 1.1: Contingent Cone when : (a) $x^\star$ is an interior point , (b) $x^\star$ is on a smooth boundary , (c) $x^\star$ is on a non smooth boundary , (d) for two distinct points $x_1^\star$ and $x_2^\star$ of the intersection $S_1 \cap S_2$

We see that $\forall x \in Int(K), \quad T_K(x) = X$. Therefore , when K is open , the contingent cone to K at any point $xinK$ is always equal to the whole space. Therefore we will be mostly interested in defining the contingent cone for point in the boundary of K . In any case the contingent cone of a point x in the boundary can be the whole space

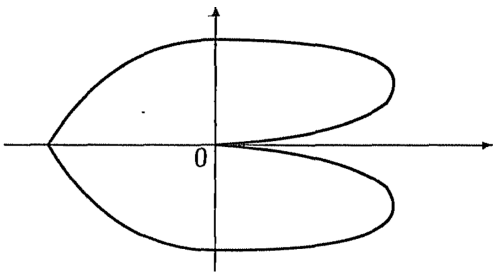


Figure 1.2: Insieme K tale che $T_k(0) = X$

Lemma 1.1.1. Let $x : [0, T] \to K$ be a differentiable viable function

$$\forall t \in [0, T) \quad x'(t) \in T_K(x)$$

Definition 1.1.4 (Viability Domain)**.** Let K be a subset of Ω. We shall say that K is a viability domain of the map $f : \Omega \to X$ se

$$\forall x \in X \quad , \quad f(x) \in T_K(x)$$

Theorem 1.1.1 – Nagumo

Let us assume that

1. K is locally compact
2. f is continuous from K to X

The K is locally viable under f if and only if K is viability domain of f

Esempio 1.1.1. Let $K \subset \mathbb{R}$, with $K = [-2, 2]$, and $\dot{x}(t) = x(t)$. K is a bounded and closed interval, therefore K is compact. Moreover, the function is continuous, therefore the hypotheses of the theorem are satisfied. Let's check whether K is a *viability domain*. We know that if $x \in (-2, 2)$, then $T_K(x) = \mathbb{R}$. Now let's analyze the two boundary points:

- ⋆ If $x = 2 \to f(x) = 2$. The tangent cone represents the directions in which I can move without exiting K. In this case, I can only move to the left, so I obtain $T_K(2) = (-\infty, 0]$.
- ⋆ Following a reasoning similar to the previous one, We obtain $T_K(-2) = [0, +\infty)$.

We can see that for both boundary points we obtain $f(x) \notin T_K(x)$, therefore K is not a viability domain for f.

Now, let's consider the interval $K = (-\infty, 0]$. In this case, the subset is not compact but locally compact, since it is a closed subset of a locally compact space, the Euclidean space. As before, if we take $x \in (-\infty, 0)$, we obtain $T_K(x) = \mathbb{R}$. Let's now move on to study the point $x = 0$, for which we obtain $f(0) = 0$; instead, the tangent cone at the point zero is $T_K(0) = (-\infty, 0]$. We can observe that $f(0) \in T_K(0)$, therefore K is a viability domain, and hence locally viable.

Esempio 1.1.2. Let $K \subset \mathbb{R}^2$ be given by $K = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 - 5 \leq 0\} = \{(x, y) \in \mathbb{R}^2 \mid g(x, y) \leq 0\}$, and let us consider the vector field $f(x, y) = (x, y^2)$. K is a compact subset and f is a continuous vector field.

We know that for interior points $x \in K^\circ$, the tangent cone is $T_K(x) = \mathbb{R}^2$. Let us now consider the boundary points $x \in \partial K$, which belong to the level curve $g^{-1}(0) = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 5\}$. In this case, the boundary curve is differentiable, and we can define the tangent cone using tangent lines. Indeed,

$$T_K(x_0) = \{v \in \mathbb{R}^2 \mid g(x_0) + \nabla g(x_0) \cdot v \leq 0\}$$
$$T_K(x_0) = \{v \in \mathbb{R}^2 \mid \nabla g(x_0) \cdot v \leq 0\}$$

Let us compute the condition for the vector field to belong to the tangent cone at the boundary:

$$\begin{aligned} \nabla g(x) \cdot f(x) &= (2x, 2y) \cdot (x, y^2) \\ &= 2x^2 + 2y^3 \leq 0 \\ &\implies y \leq -x^{\frac{2}{3}} \end{aligned}$$

Indeed, evaluating this at the boundary point $x_0 = (0, \sqrt{5})$:

$$\begin{aligned} f(x_0) &= (0, 5) \\ g(x_0) &= 0 \\ \nabla g(x_0) &= (0, 2\sqrt{5}) \\ \nabla g(x_0) \cdot f(x_0) &= 10\sqrt{5} > 0 \implies f(x_0) \notin T_K(x_0) \end{aligned}$$

We have thus shown that K is not a viability domain for f, meaning K cannot be locally viable under f.

Now, consider the same set K but with the vector field $f(x, y) = (-x, -y^3)$. Let us check if this vector field satisfies the condition for belonging to the tangent cone:

$$\begin{aligned} \nabla g(x) \cdot f(x) &= (2x, 2y) \cdot (-x, -y^3)^T \\ &= -2x^2 - 2y^4 \leq 0 \quad \forall (x, y) \in \mathbb{R}^2 \end{aligned}$$

We have therefore shown that $f(x) \in T_K(x)$ for all $x \in K$. Thus, K is a viability domain for f, and by Nagumo's Theorem, K is locally viable under f.

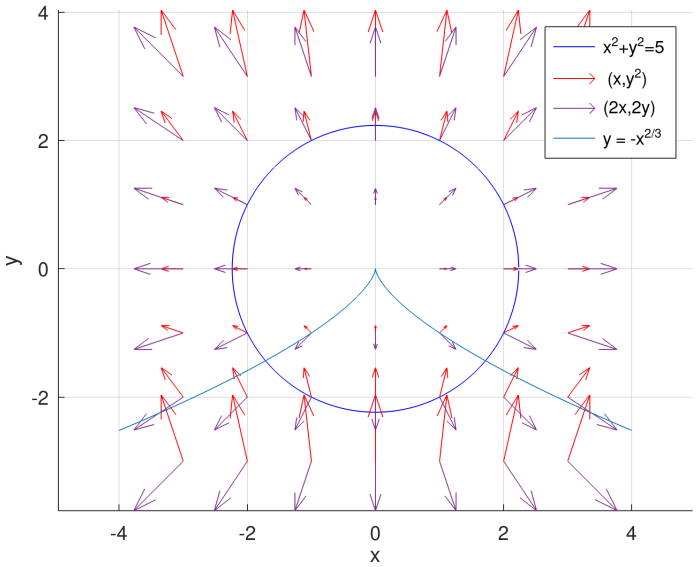


Figure 1.3: Example 1.1.2(a)

Theorem 1.1.2 – Viability

Let us consider a closed subset K of a finite dimensional vector-space X and a continuous map $f : K \to X$.
If K is a viability domain , then for every initial state $x_0 \in K$ there exists a positive T and a viable solution on $[0, T]$ to the differential equation 1.1.1 starting at x_0 such that

$$\begin{cases} T = +\infty \\ T < +\infty \quad \textit{and} \quad \limsup_{t \to T_-} \|x(t)\| = \infty \end{cases}$$

Further adequate information - a priori estimates on the growth of f - allows us to exclude the case when $\lim_{t \to T^-} \|x(t)\| = \infty$.

Theorem 1.1.3

Let us consider a subset of K of a finite dimensional vector-space X and a map f form K to X . We assume that the map f is continuous from K to X , that

$$\exists c > 0 \ t.c \ \forall x \in K \ , \ \|f(x)\| \leq c(\|x\| + 1)$$

and that K is a closed viability domain of f .
Then K is viable under f : for every initial state $x_0 \in K$, there exists a viable solution on $[0, \infty]$ to the differential equation 1.1.1 at x_0

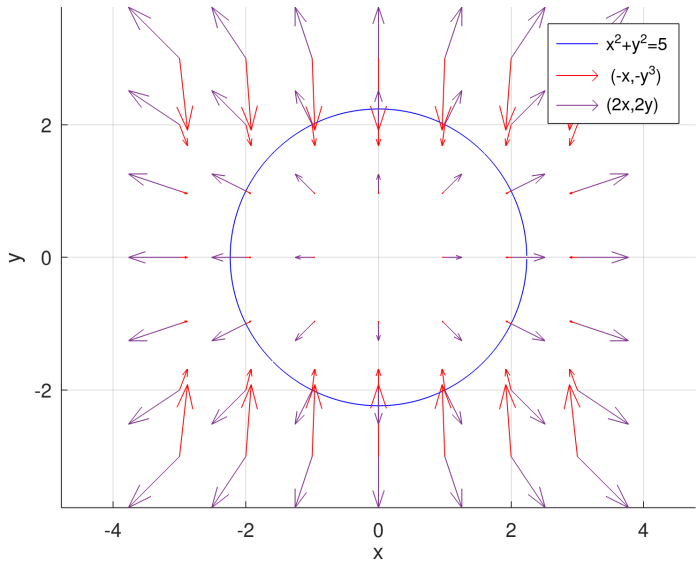


Figure 1.4: Example 1.1.2 (b)

1.2 cbf

Control barrier function are inspired by the control Lyapunov function , which are the extension to control theory of the Lyapunov function. In the same way CBF are the extension to contro theory of the barrier function , used in optimization . The main objective of CBF is to define a forward invariant subset , a subset for which every trajectory that start in the subset is confined in there for all future times . CBF are really useful in the synthesis of controls that let us stay in the foward invariant subset .
Let us consider the non-linear control affine system

$$\dot{x} = f(x) + g(x)u \tag{1.2.1}$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$ are respectively the state and the input of the system. The function $f : \mathbb{R}^n \to \mathbb{R}^n$ e $g : \mathbb{R}^n \to \mathbb{R}^{n\times m}$ are locally Lipschitz .
Let us consider the set S , defined as the superlevel set of a continuously differentiable function $h : \mathbb{R}^n \to \mathbb{R}$:

$$S = \{x \in \mathbb{R}^n | h(x) \geq 0\} \tag{1.2.2}$$
$$S^0 = \{x \in \mathbb{R}^n \mid h(x) > 0\} \tag{1.2.3}$$
$$\partial S = \{x \in \mathbb{R}^n \mid h(x) = 0\} \tag{1.2.4}$$

Definition 1.2.1. The set S is forward invariant if for every $x_0 \in S, x(t) \in S$ for $x(0) = x_0$ and all $t \in I(x_0)$. The system 1.2.1 is safe with respect to the set S if S is foward invariant

Definition 1.2.2. The $h \in \mathbb{C}^1(D), h : D \subseteq \mathbb{R}^n \to \mathbb{R}$ is control barrier function , for the system 1.2.1 , on the set S , defined by 1.2.2 - 1.2.4 , if there exists an extended class κ_∞ function α [1] such that :

$$\begin{aligned} &\sup_{u\in\mathbb{R}^m} [L_f h(x) + L_g h(x)u \geq -\alpha(h(x))] \quad \forall x \in S \\ &\sup_{u\in\mathbb{R}^m} [\boldsymbol{\nabla} h(x) \cdot f(x) + \boldsymbol{\nabla} h(x)^T g(x)u \geq -\alpha(h(x))] \quad \forall x \in S \end{aligned} \tag{1.2.5}$$

where

$$L_g h = \begin{bmatrix} L_{g_1} h \\ L_{g_2} h \\ \vdots \\ L_{g_m} h \end{bmatrix}^T$$

so g_i is the i column of g

Let us consider the set of all the control that render the set S forward invariant :

$$K_{cbf}(u) = \{u \in \mathbb{R}^m \mid L_f h(x) + L_g h(x)u \geq -\alpha(h(x))\}$$

[1] For $a \in \mathbb{R}^+$, a continuous function $\alpha : [0, a) \to \mathbb{R}$ is a class κ function if it is strictly increasing and $\alpha(0) = 0$. A continuous function $\alpha : (-b, a) \to \mathbb{R}$, with $a, b \in \mathbb{R}^+$, is an extended class κ function if it is strictly increasing and $\alpha(0) = 0$. Furthermore, if $a = b = \infty$ and $\lim_{r\to\infty} \alpha(r) = \infty$, then it is called an extended class κ_∞ function.

Theorem 1.2.1

Let $S \subset \mathbb{R}^n$ be a set defined as the superlevel set of a continuously differentiable function $h : D \subset \mathbb{R}^n \to \mathbb{R}$, if h is a control barrier function on D and $\boldsymbol{\nabla} h(x) \neq 0 \;\; \forall x \in \partial S$, the any Lipschitz continuous controller $u(x) \in K_{cbf}$ for the system 1.2.1 renders the set S forward invariant

Proof. For all $x \in \partial S$ we have $L_f h(x) + L_g h(x) u \geq -\alpha(h(x)) = 0$.Then we can define the contingent cone $T_S(x) = \{v \in \mathbb{R}^n \mid \boldsymbol{\nabla} h(x) \cdot v \geq 0\}$. Wee know that $\forall x \in S^0 \;\; T_S(x) = \mathbb{R}^n$, let's what happens for the boundary points

$$\begin{aligned} \boldsymbol{\nabla} h(x)(f(x) + g(x)u) &\geq 0 \\ \boldsymbol{\nabla} h(x) f(x) + \boldsymbol{\nabla} h(x) g(x) u &\geq 0 \\ L_f h(x) + L_g h(x) u &\geq 0 \end{aligned}$$

then S is viability domain and then S is also locally viable under f.
We are working under the hypothesis that f, g, u are locally Lipschitz function so the Cauchy-Lipschitz Theorem guarantees existence and uniqueness of the solution . Knowing this the condition for invariance and viabilitlity corrispondes , so the set S is forward invariant □

Furthermore control barrier function give us a necessary and sufficient condition for forward invariance if the set S is compact

Theorem 1.2.2

Let S be the compact superlevel set of the function $h \in C^1$, $h : \mathbb{R}^n \to \mathbb{R}$, with $\boldsymbol{\nabla} h(x) \neq 0 \;\; \forall x \in \partial S$.If there exists a control law $u = k(x)$ that renders the subset S forward invariant , then $h|_S : S \to \mathbb{R}$ è is a control barrier function on S

Proof. Let $h \in C^1, h|_S : S \to \mathbb{R}$. For all $r \geq 0$ the set $\{x \mid 0 \leq h(x) \leq r\}$ is a compact subset of C . We define the function $\alpha : [0, \infty) \to \infty$ with

$$\alpha(r) = - \inf_{\{x \mid 0 \leq h(x) \leq r\}} L_f h + L_g h \; u$$

. Using the compactness of the subset and the continuity of $L_f h + L_g h \; u$, α is well defined function , increasing and satisfies

$$L_h f + L_g h \; u \geq -\alpha(h(x))$$

For Nagumo Theorem 1.1 the forward invariance of C is equivalent to :

$$L_h f + L_g h \; u \geq 0 \qquad \forall x \in \partial S$$

, this implies that $\alpha(0) \leq 0$. We can always choose a function of class κ $\hat{\alpha}$ che maggiora α , facendo così abbiamo ottenuto $L_h f + L_g h \; u \geq -\hat{\alpha}(h(x))$. □

Now we know that the CBF give us conditions for the safety of a set , we want to find a method to creates the control laws that achieves safety .
Suppose that we have a control law u for the system 1.2.1 and we want to make sure that it enforce forward invariance . Let us suppose that $u \in K_{cbf}$ per some

$x \in \mathbb{R}^n$. We want to modify this controller to ensure forward invariance , we start by noting that the condition 1.2.5 is affine in u , then we can consider the quadratic programming problem (QP)

$$u(x) = \underset{u \in U \subseteq \mathbb{R}^n}{\operatorname{argmin}} \frac{1}{2}\|u - k(x)\|_2^2$$
$$s.t \quad [L_f h(x) + L_g h(x) u \geq -\alpha(h(x))$$

For this for this problem there exists a solution u^* in closed form :

$$u^* = u_{safe} + u_{nom} \tag{1.2.6}$$

where

$$u_{safe} = \begin{cases} -\frac{\Phi(x,t,u_{des})}{L_g h(x) L_g h(x)^T} L_g h(x)^T & \Phi < 0 \\ 0 \qquad \Phi \geq 0 \end{cases} \tag{1.2.7}$$

with

$$\Phi(x, t, u_{des}) \triangleq \boldsymbol{\nabla} h(x)^T f(x) + \boldsymbol{\nabla} h(x)^T g(x) u + \alpha(h(x)) \tag{1.2.8}$$

We now show an example of the usage of CBF

Esempio 1.2.1. We want to move a material point from an initial position $q_0 \in \mathbb{R}^2$ to a target position $q_g \in \mathbb{R}^2$ while avoiding a spherical obstacle , with center $q_{obs} \in \mathbb{R}^2$ and radius r_{obs} . The system is a single integrator :

$$\dot{q}(t) = u(t)$$

then $f(q) = 0$, $g(q) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$

We start by introducing the control barrier function

$$h(x) = \|q - q_{obs}\| - r_{obs}$$

which defines the set

$$\begin{aligned} S &= \{q \in R^2 \mid \|q - q_g\| - r_{obs} \geq 0\} \\ &= \{q \in R^2 \mid \sqrt{(x - x_{obs})^2 + (y - y_{obs})^2} \geq r_{obs}\} \end{aligned}$$

We have out nominal control which is

$$u_{nom} = K * \frac{q_g - q}{||q_g - q||} \quad K \in \mathbb{R}_{>0}$$

by using equation 1.2.8 and using a constant $\alpha \in \mathbb{R}_{>0}$ as the κ_∞ function we obtain :

$$\begin{aligned} \nabla h &= \begin{bmatrix} \frac{x}{(\sqrt{x - x_{obs}})^2 + (y - y_{obs})^2} \\ \frac{t}{\sqrt{(x - x_{obs})^2 + (y - y_{obs})^2}} \end{bmatrix} \\ L_g h &= \nabla h^T \\ L_g h L_g h^T &= \frac{x^2 + y^2}{(x - x_{obs})^2 + (y - y_{obs})^2} \\ \Phi &= \frac{x}{(\sqrt{x - x_{obs})^2 + (y - y_{obs})^2}} u_1 + \frac{y}{(\sqrt{x - x_{obs})^2 + (y - y_{obs})^2}} u_2 + \alpha(\|q - q_{obs}\| - r_{obs}) \end{aligned}$$

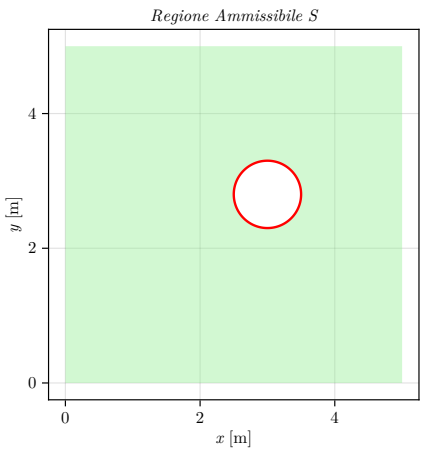
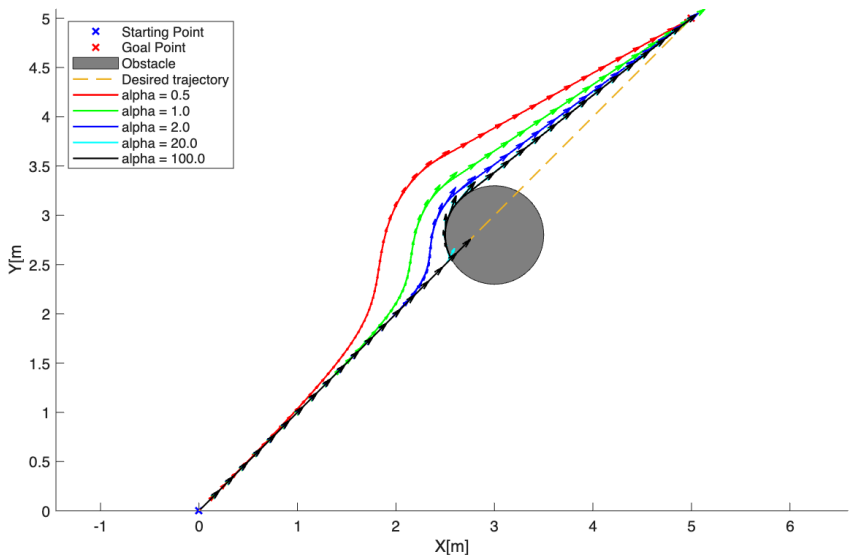


Figure 1.5: The set S

and now we have u_{safe} and by adding out u_{nom} we can create our controller that achieves obstacle avoidance while going toward the goal .



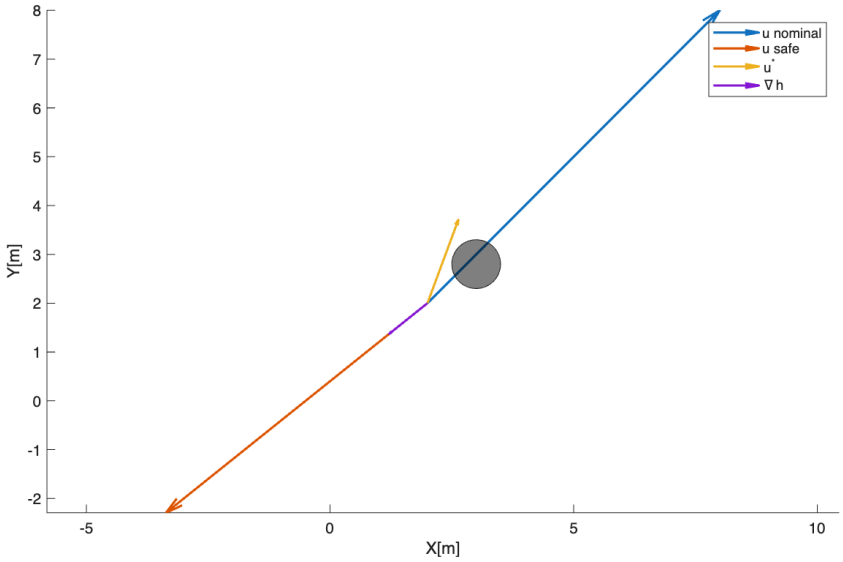
In the next figure we will see how every component $(u^*, u_{safe}, u_{nom}, \boldsymbol{\nabla} h)$ is for the point $q = (2, 2)$

1.2.1 Limbo Problem

Equazioni del moto

$$\begin{cases} ml^2\ddot{\theta} - m\ddot{x}l\cos(\theta) - mgl\sin(\theta) = 0 \\ (M+m)\ddot{x} - ml\ddot{\theta}\cos(\theta) + ml\dot{\theta}^2\sin(\theta) = u \end{cases}$$

$$\begin{cases} \ddot{x} = \frac{u+m\sin(\theta)\left(g\cos(\theta)-l\dot{\theta}^2\right)}{M+m\sin^2(\theta)} \\ \ddot{\theta} = \frac{(M+m)g\sin(\theta)-ml\dot{\theta}^2\sin(\theta)\cos(\theta)+u\cos(\theta)}{l\left(M+m\sin^2(\theta)\right)} \end{cases}$$



We find that the equilibrium point are in the form $(x, \theta, u) = (x, 0, 0)$

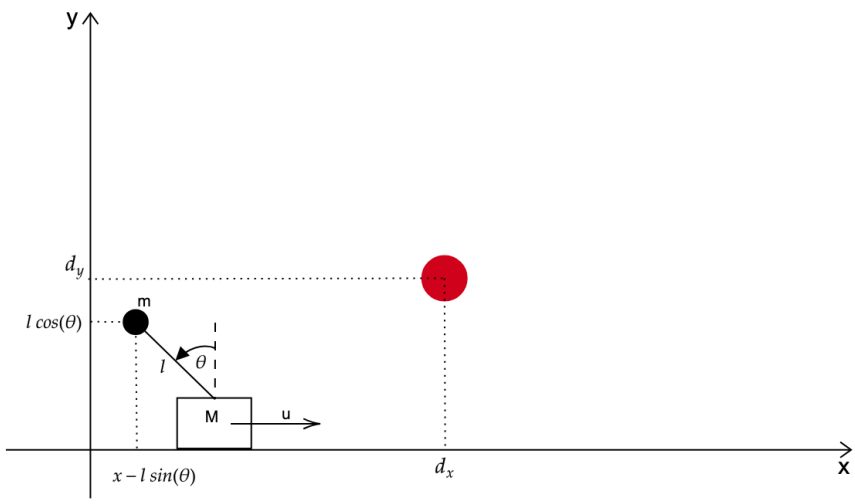


Figure 1.6: Schema del sistema carrello-pendolo inverso.

Angolo Critico

After some calculations we obtain :

$$\ddot{\theta} = \frac{-m\cos(\theta)\sin(\theta)\dot{\theta}^2}{(M + m\sin(\theta)^2} + u\frac{\cos(\theta)}{l(M + m\sin(\theta)^2)} + \frac{(M+m)g\sin(\theta)}{l(M + m\sin(\theta)^2)}$$

We can rewrite by defining the function

$$g(\theta) = \frac{\cos(\theta)}{l(M + m\sin(\theta)^2)}$$
$$h(\theta, \dot{\theta}, u) = lm\dot{\theta}^2 - \frac{u + g(M+m)\tan(\theta)}{\sin(\theta)}$$
$$\ddot{\theta} = -\sin(\theta)g(\theta)h(\theta, \dot{\theta}, u)$$

We want to prove that there exist a crtical anglie $\theta_c \in]-\pi/2, \pi/2[$. We note that $g(\theta) > 0 \quad \forall\theta \in]-\pi/2, \pi/2[$, so if we want the acceleration the have the opposite sign of θ we need to guarantee that $h(\theta, \dot{\theta}, u) \geq 0$

$$lm\dot{\theta}^2 - \frac{u + g(M+m)\tan(\theta)}{\sin(\theta)} \geq 0$$
$$lm\dot{\theta}^2 \geq \frac{u + g(M+m)\tan(\theta)}{\sin(\theta)}$$
$$\text{Now we take the Infimum}(\dot{\theta} = 0)$$
$$\frac{u + g(M+m)\tan(\theta)}{\sin(\theta)} \leq 0$$

We now have two cases

- $\star$ $\sin(\theta) > 0$

$$u_{min} \leq u \leq -g(M+m)\tan(\theta)$$
$$\tan(\theta) \leq \frac{-u_{min}}{g(M+m)}$$

- $\star$ $\sin(\theta) < 0$

$$u_{max} \geq u \geq -g(M+m)\tan(\theta)$$
$$\tan(\theta) \geq \frac{-u_{max}}{g(M+m)}$$

so we obtain

$$\arctan\left(-\frac{u_{max}}{g(M+m)}\right) \leq \theta \leq \arctan\left(-\frac{u_{min}}{g(M+m)}\right)$$

1.3 Geometric Control Theory

Definition 1.3.1. The system 1.2.1 is said to be controllable if for any two points x_0 and x_f in M there exist and admissible controlo $u(t)$ defined on some time interval $[0, T]$ such that the system 1.2.1 with initial condition x_0 reaches the point x_f in time T

Generally it's not easy to prova that a system is contrabble , so we introduce a related property called **accessability** , whisch is often much easier to prove . We first need the notion of reachable set

Definition 1.3.2. Given $x_0 \in M$ we define $R(x_0, t)$ to be the set of all $x \in M$ for which there exists an admissinle controlo u such taht there is a trajectory of 1.2.1 with $x(0) = x_0 \quad , \quad x(t)) = x$. The reachable set form x_0 at time T is definied to be

$$R_T(x_0) = \bigcup_{0 \leq t \leq T} R(x_0, t)$$

Definition 1.3.3. The accessibility algebra C of 1.2.1 is the smallest Lie Algebra of vector field onb M that constains the vector Fields f and $g_1, \ldots, g_m$

The Accessibility algebra is just the span of all possible Lie brackets of f and the g_i

Definition 1.3.4. We define the accessibility distribution C of 1.2.1 to be the distribution generated by the vector field in C , i.e. , $C(x)$ is the span of the vector field X in C at z

Definition 1.3.5. The system 1.2.1 is said to be accessible from $p \in M$ if for evey $T > 0$, $R_T(p)$ contains a non empty open set

Theorem 1.3.1

Consider the system 1.2.1 and assume that the vector field are C^∞. If $\dim C(x_0) = n$ (the accessibility algebra span the tangent space to M at x_0) the for any $T > 0$ the set $R_t(x_0)$ has a nonempry interior ; i.e. , the system ha the accessibility propery from x_0

Definition 1.3.6. A **system** or **machine** $\Sigma = (\mathcal{T}, X, \mathcal{U}, \phi)$ consists of:

- ⋆ A time set $\mathcal{T}$;
- ⋆ A nonempty set X called the **state space** of Σ;
- ⋆ A nonempty set $\mathcal{U}$ called the **control-value** or **input-value space** of Σ; and
- ⋆ A map $\phi : \mathcal{D}_\phi \to X$ called the **transition map** of Σ, which is defined on a subset $\mathcal{D}_\phi$ of

$$\{(\tau, \sigma, x, \omega) \mid \sigma, \tau \in \mathcal{T}, \sigma \leq \tau, x \in X, \omega \in \mathcal{U}^{[\sigma,\tau)}\}, \tag{1.3.1}$$

such that the following properties hold:

nontriviality For each state $x \in X$, there is at least one pair $\sigma < \tau$ in $\mathcal{T}$ and some $\omega \in \mathcal{U}^{[\sigma,\tau)}$ such that ω **is admissible for** x, that is, so that $(\tau, \sigma, x, \omega) \in \mathcal{D}_\phi$;

restriction If $\omega \in \mathcal{U}^{[\sigma,\mu)}$ is admissible for x, then for each $\tau \in [\sigma, \mu)$ the restriction $\omega_1 := \omega|_{[\sigma,\tau)}$ of ω to the subinterval $[\sigma, \tau)$ is also admissible for x and the restriction $\omega_2 := \omega|_{[\tau,\mu)}$ is admissible for $\phi(\tau, \sigma, x, \omega_1)$;

semigroup If σ, τ, μ are any three elements of $\mathcal{T}$ so that $\sigma < \tau < \mu$, if $\omega_1 \in \mathcal{U}^{[\sigma,\tau)}$ and $\omega_2 \in \mathcal{U}^{[\tau,\mu)}$, and if x is a state so that

$$\phi(\tau, \sigma, x, \omega_1) = x_1 \quad \text{and} \quad \phi(\mu, \tau, x_1, \omega_2) = x_2,$$

then $\omega = \omega_1\omega_2$ is also admissible for x and $\phi(\mu, \sigma, x, \omega) = x_2$;

identity For each $\sigma \in \mathcal{T}$ and each $x \in X$, the empty sequence $\diamond \in \mathcal{U}^{[\sigma,\sigma)}$ is admissible for x and $\phi(\sigma, \sigma, x, \diamond) = x$. □

Definition 1.3.7. Let $\mathcal{X}$ be an open subset of $\mathbb{K}^n$ and let $\mathcal{U}$ be a metric space. A **right hand side** with respect to $\mathcal{X}$ is a function

$$f : \mathbb{R} \times \mathcal{X} \times \mathcal{U} \to \mathbb{K}^n$$

which can be obtained in the following way : There must exist another metric space S as well as maps

$$\tilde{f} : S \times \mathcal{X} \times \mathcal{U} \to \mathbb{K}^n$$

and

$$\pi : \mathbb{R} \to S$$

so that

$$f(t, x, u) = \tilde{f}(\pi(t), x, u)$$

and the following properties hold :

- ⋆ $\tilde{f}(s, \cdot, u)$ is of class C^1 for each fixed s,u
- ⋆ both $\tilde{f}$ and the partial derivative $\tilde{f}_x$ are continuos on $S \times \mathcal{X} \times \mathcal{U}$
- ⋆ π is a mueasurable locally essentially bounded function

For f independent of t , the definition reduces to the requirement that U be a metric space and

$$f : \mathcal{X} \times \mathcal{U} \to \mathbb{K}^n$$

and

$$f \quad \text{and} \quad f_x \quad \text{are continuos on} \quad \mathcal{X} \times \mathcal{U}$$

Definition 1.3.8. Let $k = 1, 2, \ldots, \infty$. A **continuos-time system of class** $C^(k)$ is one for which U is an open subset of K^m , for some nonnegative integer m , and for which S and $\tilde{f}$ can be chosen so that S is an open subset of a Euclidean space and $\tilde{f}$ is of class C^k

Proposizione 1.3.1. if Σ is either a continuous-time system , or a discrete system of class C^0 , then Σ is topological when $\mathcal{X}$ is given the induced topologu of $\mathbb{R}^n$

Definition 1.3.9. The system Σ is locally controllable in time T at the equilibrium state x if and only if for every neighborhood $\mathcal{V}$ of x there is another neighboord $\mathcal{W}$ of x , such that , for any pair of elements z , y in W , z can be controlled to y inside $\mathcal{V}$ in time T

Recall the notion of equilibrium state $x \in \mathcal{X}$; for a time-invariant system, this is equivalent to the requirement that there exists a $u \in \mathcal{U}$ such that for all $T \in \mathcal{T}_+$, $\phi(T, 0, x, \omega) = x$, where $\omega \equiv u$. For discrete-time systems this is equivalent to

$$\mathcal{P}(x, u) = x$$

and for continuous-time systems to

$$f(x, u) = 0\,.$$

Theorem 1.3.2

Assume that Σ is a continuos-time system of class C^1 . Then ,

1. Let $\Gamma = (\xi, \omega)$ be a trajectory for Σ on an interval $I = [\sigma, \tau]$. Thean , a sufficient condition for Σ to be locally controllable along ξ is that $\sigma_*[\Gamma]$ be controllable on $[\sigma, \tau]$
2. Ler Σ be time-invariant , and assume that x is an equilibrium state . Pick ant $\varepsilon > 0$ and any $u \in U$ such that $f(x, u) = 0$. Then , a sufficient condition for Σ to be locally controllable at in time ϵ is that the linearization of Σ at (x,u) be controllable

We now define the controllabilty of a linear sisystem

Theorem 1.3.3

The n-dimensionale continuos and invariant time linear system Σ is controllable if and only if $rankR(A, B) = n$ where

$$R(A, B) = [B, AB, A^2B, \ldots, A^{n-1}B]$$

1.3.1 Inverse Pendulum

We denoted with $P = (P_x, P_y) = (x - l\sin(\theta), l\cos(\theta))$ the coordinates of the pendulum and with $d = (d_x, d_y)$ the coordinates of the center of the obstacle of radius ϵ . The state vector is $z = [x, \theta, \dot{x}, \dot{\theta}]$

$$\begin{aligned}
h_0(z) &= (x - l\sin(\theta) - d_x)^2 + (l\cos(\theta) - d_y)^2 - \epsilon^2 \\
&= x^2 + l^2\sin^2(\theta) - d_x^2 - 2xl\sin(\theta) - 2xd_x + 2d_xl\sin(\theta) + \\
&+ l^2\cos^2(\theta) + d_y^2 - 2d_yl\cos(\theta) - \epsilon^2 \\
&= x^2 + l^2 + d_x^2 + d_y^2 - 2(xl\sin(\theta) + xd_x - d_xl\sin(\theta) + d_yl\cos(\theta)) - \epsilon^2
\end{aligned}$$

We compute

$$\nabla h_0(z) = \begin{bmatrix} 2x - 2lsin(\theta) - 2d_x \\ -2(xl\cos(\theta) - d_xl\cos(\theta) - dyl\sin(\theta)) \\ 0 \\ 0 \end{bmatrix}$$

Then we need to compute the term

$$\nabla h_0 \cdot (f + gu) =$$
$$= \begin{bmatrix} 2x - 2lsin(\theta) - 2d_x \\ -2(xl\cos(\theta) - d_x l\cos(\theta) - dyl\sin(\theta)) \\ 0 \\ 0 \end{bmatrix} \cdot \left(\begin{bmatrix} v_x \\ v_\theta \\ \frac{m\sin(\theta)(g\cos(\theta) - lv_\theta^2)}{M + m\sin^2(\theta)} \\ \frac{(M+m)g\sin(\theta) - mlv_\theta^2\sin(\theta)\cos(\theta)}{l(M + m\sin^2(\theta))} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{M + m\sin^2(\theta)} \\ \frac{\cos(\theta)}{l(M + m\sin^2(\theta))} \end{bmatrix} u \right)$$
$$= v_x(2x - 2lsin(\theta) - 2d_x) - 2v_\theta(xl\cos(\theta) - d_x l\cos(\theta) - dyl\sin(\theta))$$

and we obtain

$$\begin{aligned} h_1 &= \dot{h_0} + k_1 h_0 \quad k_1 \in \mathbb{R} \\ &= \nabla h_0 \cdot (f + gu) + k_1 h_0 \\ &= v_x(2x - 2lsin(\theta) - 2d_x) - 2v_\theta(xl\cos(\theta) - d_x l\cos(\theta) - dyl\sin(\theta)) + \\ &+ k_1(x^2 + l^2 + d_x^2 + d_y^2 - 2(xl\sin(\theta) + xd_x - d_x l\sin(\theta) + d_y l\cos(\theta))) - \epsilon^2 \end{aligned}$$

Now we define the safe set as $S = S_1 \cap S_2$ where

$$\begin{aligned} S_1 &= \{z \in \mathbb{R}^4 \mid h_0(z) \geq 0\} \\ &= \{z \in \mathbb{R}^4 \mid x^2 + l^2 + d_x^2 + d_y^2 - 2(xl\sin(\theta) + xd_x - d_x l\sin(\theta) + d_y l\cos(\theta)) \geq \epsilon^2\} \end{aligned}$$

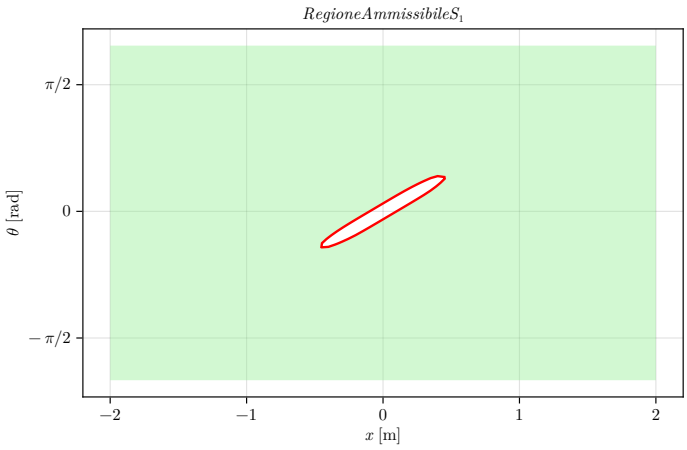


Figure 1.7: The safe set S_1 colored in green

$$\begin{aligned} S_2 &= \{z \in R^4 \mid h_1(z) \geq 0\} \\ &= \{z \in R^4 \mid v_x(2x - 2lsin(\theta) - 2d_x) - 2v_\theta(xl\cos(\theta) - d_x l\cos(\theta) - d_y l\sin(\theta)) + \\ &+ k_1(x^2 + l^2 + d_x^2 + d_y^2 - 2(xl\sin(\theta) + xd_x - d_x l\sin(\theta) + d_y l\cos(\theta))) \geq k_1\epsilon^2\} \end{aligned}$$

We also have defined another control barrier function $b_0(z) = \theta_{cri}^2 - \theta^2$. We

need to compute the gradient and the term $dot(b_0)$in this case

$$\nabla b_0 = \begin{bmatrix} 0 \\ -2\theta \\ 0 \\ 0 \end{bmatrix}$$

$$\nabla b_0 \cdot (f + gu) =$$

$$= \begin{bmatrix} 0 \\ -2\theta \\ 0 \\ 0 \end{bmatrix} \cdot \left(\begin{bmatrix} v_x \\ v_\theta \\ \frac{m\sin(\theta)(g\cos(\theta) - lv_\theta^2)}{M + m\sin^2(\theta)} \\ \frac{(M+m)g\sin(\theta) - mlv_\theta^2\sin(\theta)\cos(\theta)}{l(M + m\sin^2(\theta))} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{M + m\sin^2(\theta)} \\ \frac{\cos(\theta)}{l(M + m\sin^2(\theta))} \end{bmatrix} u \right)$$

$$= -2v_\theta\theta$$

Then we can define

$$b_1 = -2v_\theta\theta + alpha_1(\theta_{cri}^2 - \theta^2)$$

The safe set in this caso is given by $B = B_1 \cap B_2$

$$B_1 = \{z \in \mathbb{R}^4 \mid \theta_{cri}^2 \geq \theta^2\}$$

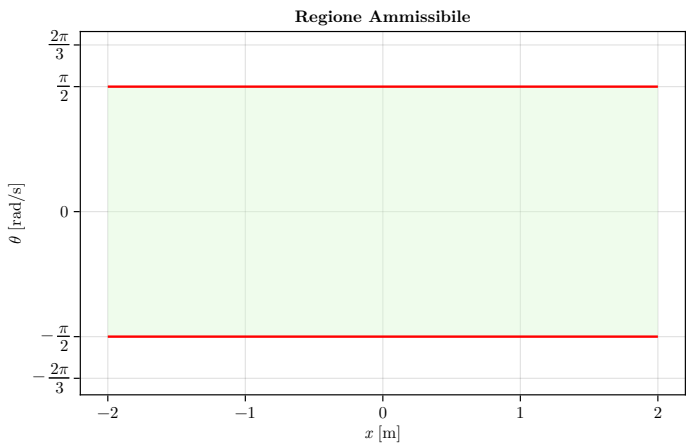


Figure 1.8: The set B_1 colored in green

$$B_2 = \{z \in \mathbb{R}^4 \mid -2v_\theta\theta + \alpha_1(\theta_{cri}^2 - \theta^2) \geq 0\}$$

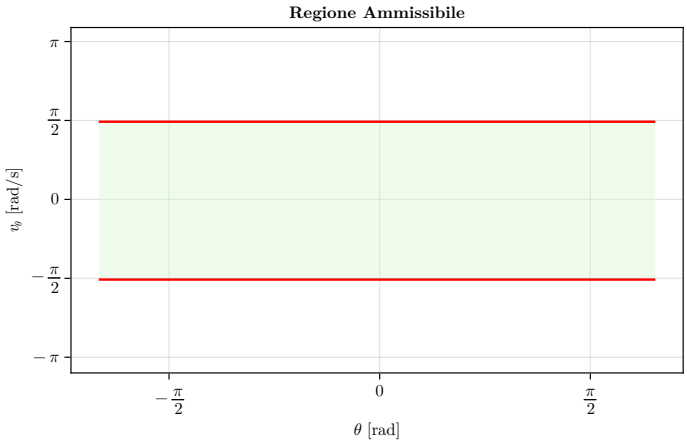


Figure 1.9: The set B_2 colored in green with $\alpha_1 = 40$

The Final boss is $C = S \cap B$ which i refused to write

.1 Richiami di Topologia

Definition .1.1 (Spazio topologico)**.** Sia X un insieme , una *topoogia* su X , è una famiglia τ di sottoinsiemi di X (i suoi elementi sono gli aperti di X) che soddisfa le seguenti condizioni.

- $\star$ $\emptyset$ *e* $X \in \tau$
- $\star$ Unione arbitraria di aperti è un sottoinsieme aperto (se $A_\lambda \in \tau$ per ogni $\lambda \in \Lambda$, allora $\bigcup_{\lambda \in \Lambda} A_\lambda \in \tau$)
- $\star$ Intersezione finita di aperti è un sottoinsieme aperto (Se $A_1, \ldots, A_m \in \tau$ allora $A_1 \cap \cdots \cap A_m \in \tau$0-)

Un insieme dotato di una topologia viene detto **spazio topologico**

Esempio .1.1. Su ogni insieme X , $\tau = \{\emptyset, X\}$ è una topologia detta banale od indiscreta.
Sull'insieme $\mathbb{R}$

$$\tau_i = \{\emptyset, \mathbb{R}\} \cup \{\ ]a, +\infty[\ \}$$

è una topologia , topologia inferiore , similmente

$$\tau_\sigma = \{\emptyset, \mathbb{R}\} \cup \{\ ]-\infty, a[\ \}$$

e1 la topologia superiore di $\mathbb{R}$

Una descrizione esplicita di tutti gli aperti di uno spazio topologico e1 impossibile , la topologia viene in genere descritta assegnando una base per essa

Definition .1.2. Sia τ una topologia su insieme X . una sottofamiglia (un insieme) $B \subset \tau$ si dice una base di τ se ogni aperto $A \in \tau$ può essere scritt ocome unione di elementi d iB

Teorema .1.0.1. *Sia X un insieme e $\mathcal{B} \subset P(X)$ una famiglia di suoi sottoinsiemi . Allora esiste una topologia su X di cui $\mathcal{B}$ è una base se e soltanto se sono soddisfatte le seguenti due condizioni*

- $\star$ $X = \cup\{B \mid B \in \mathcal{B}\}$
- $\star$ *Per ogni coppia $A, B \in \mathcal{B}$ e per ogni punto $x \in A \cap B$ esiste $C \in \mathcal{B}$ tale che $x \in X \subset A \cup B$*

.1.1 Parte interna , chiusura ed intorni

Definition .1.3. Sia X uno spazio topologico e $B \subseteq X$. Si denota con

- $\star$ B^0 l'unione di tutti gli aperti contenuti in B
- $\star$ $\overline{B}$ l'intersezione di tutti i chiusi contenenti in B
- $\star$ $\partial B = \overline{B} - B^0$

L'insieme B^0 viene detto parte interna di B ed è il più gtande aperto contenuto in B
L'insieme $\overline{B}$ è il più piccolo chiuso contente B e viene detto chiusura di B
Il sottoinsieme ∂B è l'intersezione dei due chiusi $\overline{B}$ e $X - B^0$ e viene detto **frontiera** di B
Osserviamo che un sottoinsieme B è aperto se e solo se $B = B^0$ e chiuse se $B = \overline{B}$

Definition .1.4. Sia X uno spazio topologico e $x \in X$. Un sottoinsieme $U \subset X$ si dice *intorno di x* se x è un punto interno di U , cioè se esiste un aperto V tale che $x \in V$ e $V \subset U$

Indichiamo con $\mathcal{I}(x)$ la famiglia di tutti gli intorni di x . per definizione se A è un sottoinsieme di uno spazio topologico , allora $A^0 = \{x \in A | A \in \mathcal{I}(x)\}$

Definition .1.5. Sia x un punto di uno spazio topologico X. Un sottofamiglia $\mathcal{J} \subset \mathcal{I}(x)$ si dice *base locale* oppure un sistema fondamentale di intorni di x , se per ogni $U \in \mathcal{I}(x)$ esiste $A \in \mathcal{J}$ tale che $A \subset U$

Esempio .1.2. Sia $U \in \mathcal{I}(x)$ un intorno fissato . Allora tutti gli intorni di x contenuti in U formano un sistema fondamentale di intorni di x

.1.2 Spazi Metrici

Definition .1.6. Una distanza si di un insieme X è un 'applicazione $d : X \times X \to \mathbb{R}$ che soddisfa le seguenti proprietà :

1. $d(x, y) \geq 0$ per ogni $x, y \in X$ e vale $d(x, y) = 0$ se e solo se x=y
2. $d(x, y) = d(y, x)$ per ogni $x, y \in X$
3. $d(x, y) \leq d(x, z) + d(z, y)$ per ogni $x, y, z \in x$ (Disuguaglianza triangolare)

Esempio .1.3. Su un qualsiasi insieme X , la funzione

$$d : X \times X \to \mathbb{R} \quad d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

è una distanza

Definition .1.7. Uno **spazio metrico** è una coppia (X, d) , dove X è un insieme e d è una distanza su X

Definition .1.8. Sia (X,d) uno spazio metrico. Il sottoinsieme

$$B(x, r) = \{y \in X | d(x, y) < r\}$$

viene detto palla paerta di centro x e raggio r

Definition .1.9 (Topologia indotta da una distanza)**.** Sia (X,d) uno spazio metrico. Nella topologia su X indotta dalla distanza d , un sottoinsieme $A \subset X$ è aperto se per ogni $x \in A$ esiste r > 0 tale che $B(x, r) \subset A$

Definition .1.10. Siano (X, d) *e* (Y, ρ) due spazi metrici , sia f una funzione $f : X \to Y$. f si dice Liptschiziana se esiste una costante $l \geq 0$ tale che sia 1

$$\rho(f(x), f(y)) \leq l \; d(x, y) \quad \forall x, y \in X$$

.1.3 Ricoprimenti

Definition .1.11. Un **ricoprimento** di un insieme X è una famiglia $\mathcal{A}$ di sottoinsieme tali che $X = \cup\{A \mid A \in \mathcal{A}\}$. diremo che il ricoprimento è finito se $\mathcal{A}$ è una famiglia finita : numerabile se $\mathcal{A}$ è una famiglia numerabile .
Se $\mathcal{A}$ *e* $\mathcal{B}$ sono ricoprimento di X se $\mathcal{A} \subset \mathcal{B}$, allora dire che $\mathcal{A}$ è un **sottoricoprimento** di $\mathcal{B}$

Definition .1.12. Un ricoprimento $\mathcal{A}$ di uno spazio topologico X si dice :

- $\star$ aperto se ogni $A \in \mathcal{A}$ è aperto
- $\star$ chiuso se ogni $A \in \mathcal{A}$ è chiuso
- $\star$ localmente finito se per ogni punto $x \in X$ esiste un aperto $V \subset X$ tale che $x \in V$ *e* $V \cap A \neq \emptyset$ per al più un numero finito di $A \in \mathcal{A}$

As esempio , ogni base della topologia è un ricoprimento aperto.

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